

### CO5 – AT3 – Level 3

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1. Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10.

**Code:**

```
import java.util.Scanner;

public class PerfectSquares {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter lower range: ");

        int l = sc.nextInt();

        System.out.print("Enter upper range: ");

        int u = sc.nextInt();

        for (int i = l; i <= u; i++) {

            int n = (int)Math.sqrt(i);

            int temp = i;

            int sum = 0;

            while (temp > 0) {

                sum = sum + temp % 10;

                temp = temp / 10;

            }

            if (n * n == i && sum < 10) {

                System.out.print(i + " ");

            }

        }

    }

}
```

```
    }  
  }  
}  
}
```

**Sample Input:**

Enter lower range: 1

Enter upper range: 40

**Sample Output:**

1 4 9 16 25 36

**2. Full Question**

Roman numerals are represented by seven different symbols: **I, V, X, L, C, D and M.**

**Symbol Value**

|   |      |
|---|------|
| I | 1    |
| V | 5    |
| X | 10   |
| L | 50   |
| C | 100  |
| D | 500  |
| M | 1000 |

For example, **2** is written as **II** in Roman numeral, just two ones added together. **12** is written as **XII**, which is simply X + II. The number **27** is written as **XXVII**, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as **IV**. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as **IX**.

There are six instances where subtraction is used:

- **I** can be placed before **V (5)** and **X (10)** to make 4 and 9.
- **X** can be placed before **L (50)** and **C (100)** to make 40 and 90.
- **C** can be placed before **D (500)** and **M (1000)** to make 400 and 900.

### Task

Given a roman numeral, convert it to an integer.

Code:

```
import java.util.Scanner;

public class Roman {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roman numeral: ");

        String s = sc.nextLine().toUpperCase();

        int result = 0;

        for (int i = 0; i < s.length(); i++) {

            int current = 0;

            int next = 0;

            char ch = s.charAt(i);

            if (ch == 'I') {

                current = 1;

            } else if (ch == 'V') {

                current = 5;
```

```
} else if (ch == 'X') {  
    current = 10;  
} else if (ch == 'L') {  
    current = 50;  
} else if (ch == 'C') {  
    current = 100;  
} else if (ch == 'D') {  
    current = 500;  
} else if (ch == 'M') {  
    current = 1000;  
}  
if (i + 1 < s.length()) {  
    char nextChar = s.charAt(i + 1);  
  
    if (nextChar == 'I') {  
        next = 1;  
    } else if (nextChar == 'V') {  
        next = 5;  
    } else if (nextChar == 'X') {  
        next = 10;  
    } else if (nextChar == 'L') {  
        next = 50;  
    } else if (nextChar == 'C') {  
        next = 100;  
    } else if (nextChar == 'D') {  
        next = 500;  
    }  
}
```

```
        } else if (nextChar == 'M') {  
            next = 1000;  
        }  
    }  
    if (current < next) {  
        result = result - current;  
    } else {  
        result = result + current;  
    }  
}  
System.out.println(result);  
}  
}
```

**Sample Input:**

Enter Roman numeral: III

**Sample Output:**

3